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Nonlinear Effects of Digital Factor Allocation on Regional Coordinated Development: Evidence from a Provincial Panel Threshold Model in China

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Abstract

China's digital economy has grown rapidly, but whether digital factor allocation promotes regional coordinated development remains an open question. This study constructs a digital factor allocation (DFA) index covering digital infrastructure, digital industry, digital finance, and digital human capital, and a regional coordinated development (RCD) index covering economic, social, and urban-rural coordination for 30 Chinese provinces from 2013 to 2021. Using the entropy weight method and a Hansen panel threshold model, this study examines the nonlinear relationship between DFA and RCD. The results show that DFA is positively associated with RCD: the baseline two-way fixed-effects estimate is 0.241 ($p < 0.05$). The association is nonlinear. When per capita GDP exceeds the estimated threshold of 10.589 (about CNY 39,700), the marginal association rises from 0.506 to 0.677. A financial development threshold indicates that the marginal association declines at very high levels of financial depth, consistent with diminishing returns to digital finance. The association is strongest in western provinces, and channel analysis suggests that industrial upgrading, financial deepening, and technological investment are plausible mechanisms. These findings imply that digital factor policies should be sequenced with economic development, tailored to regional absorptive capacity, and accompanied by complementary reforms in advanced regions.

Keywords: digital factor allocation; regional coordinated development; threshold effect; absorptive capacity; digital finance; China

1. Introduction

Over the past decade, China has built one of the world's largest digital economies. Broadband networks, mobile payments, e-commerce platforms, and cloud-based public services have expanded from leading coastal cities into inland and rural areas. This expansion has changed how firms organize production, how households access finance and markets, and how governments deliver public services (Goldfarb & Tucker, 2019). Yet the gains have not been evenly distributed. Per capita income, public-service capacity, and digital infrastructure remain uneven across provinces, and the gap between leading and lagging regions is a central concern of China's regional policy (World Bank, 2009; K. Zhang & Pei, 2019).

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Digital factor allocation (DFA)—the combined availability and application of digital infrastructure, digital industry, digital finance, and digital human capital across regions—is a potential lever for regional coordinated development (RCD). Digital infrastructure lowers the cost of market access and information diffusion; digital finance extends payment, credit, and insurance services to households and small firms that were previously excluded; digital industry creates new employment and productivity opportunities; and digital human capital determines whether these opportunities are absorbed locally (Czernich et al., 2011; Hjort & Poulsen, 2019; Jack & Suri, 2014; X. Zhang et al., 2019). In this sense, DFA can support economic coordination, social coordination, and urban-rural coordination, the three core dimensions of RCD.

The same forces can also widen regional gaps. Digital investment tends to cluster where complementary skills, finance, and infrastructure are already available (Akerman et al., 2015; Forman et al., 2012). Platform markets reward scale and concentration, and automation can displace routine labor in regions that are least able to adjust (Acemoglu & Restrepo, 2020; Autor & Salomons, 2018). Rural and lagging regions may gain access to digital services but lack the capability to convert access into productivity and income gains (Norris, 2001; van Dijk, 2005; Warschauer, 2003). Whether DFA promotes or undermines RCD is therefore an empirical question, and the answer may depend on the stage of economic development and the local absorptive capacity.

Existing research on digitalization and regional development can be grouped into three strands. The first strand focuses on measurement. Studies construct digital economy indices from internet penetration, related employment, digital output, mobile internet, and digital financial inclusion (Guo et al., 2020; Zhao et al., 2020). These indices are useful for ranking regions, but they often aggregate heterogeneous digital factors into a single score and do not distinguish infrastructure, industry, finance, and human capital. The second strand examines the effects of digitalization on growth, productivity, employment, and inclusion (Czernich et al., 2011; Hjort & Poulsen, 2019; Huang et al., 2019; X. Zhang et al., 2019). The evidence is broadly positive, but the estimated effects vary across settings and are often sensitive to the level of development. The third strand introduces nonlinearity. Threshold models show that digitalization can have stage-dependent effects, with the marginal effect changing once a critical level of digital development, market efficiency, or technological progress is reached (Qi et al., 2025). These studies suggest that linear estimates may mask important heterogeneity.

Three gaps remain. First, most studies use a composite digital economy index and do not examine whether the effect of DFA depends on complementary economic conditions. A composite index can be dominated by leading provinces and can obscure the separate roles of infrastructure, industry, finance, and human capital. Second, RCD is often measured by a single indicator, such as GDP growth or the urban-rural income ratio, rather than by a multidimensional index that reflects economic, social, and urban-rural coordination. Third, the threshold evidence for China is still limited, and the mechanisms through which DFA affects RCD are not well identified.

To fill these gaps, this study constructs a multidimensional DFA index and a multidimensional RCD index for 30 Chinese provinces from 2013 to 2021. The DFA index covers digital infrastructure, digital industry, digital finance, and digital human capital. The RCD index covers economic coordination, social coordination, and urban-rural coordination. Both indices are constructed with the entropy weight method, which avoids subjective weighting and preserves the information content of the underlying indicators. The study then applies a Hansen panel threshold model to examine whether the effect of DFA on RCD changes once economic development or financial development crosses a critical level.

The analysis also examines regional heterogeneity and three plausible channels: industrial upgrading, financial deepening, and technological investment.

The contributions of this study are fourfold. (1) **Measurement**: It develops a transparent DFA index that distinguishes four digital factor categories and an RCD index that covers three coordination dimensions, allowing the analysis to move beyond a single composite score. (2) **Nonlinear identification**: It uses a panel threshold model to test whether the DFA effect depends on economic development and financial development, and it reports bootstrap tests and likelihood-ratio confidence intervals for the threshold estimates. (3) **Regional and channel evidence**: It shows how the effect varies across regions and identifies industrial upgrading, financial deepening, and technological investment as plausible channels. (4) **Policy relevance**: It provides a differentiated policy framework that sequences digital factor allocation with economic development and complementary reforms.

The results show that DFA is positively associated with RCD. Under two-way fixed effects, a one-unit increase in the DFA index is associated with an increase of 0.241 points in the RCD index ($p < 0.05$). The association is nonlinear. When per capita GDP exceeds the estimated threshold of 10.589 (about CNY 39,700), the marginal association rises from 0.506 to 0.677. A financial development threshold indicates that the marginal association declines at very high levels of financial depth, consistent with diminishing returns to digital finance. The association is strongest in western provinces, and the channel analysis suggests that industrial upgrading, financial deepening, and technological investment are plausible mechanisms. These findings are consistent with an absorptive-capacity interpretation: digital factor allocation is most effective when local economic conditions allow digital inputs to be combined with complementary skills, markets, and institutions.

The remainder of this paper is organized as follows. Section 2 reviews the literature and develops the hypotheses. Section 3 describes the data, indices, and empirical strategy. Section 4 reports the results. Section 5 discusses the findings, policy implications, and limitations. Section 6 concludes.

2. Literature Review and Hypotheses

2.1. Digital Factor Allocation and Regional Coordinated Development

Regional coordinated development refers to the joint improvement of economic performance, social welfare, and urban-rural balance across regions. It is not simply the convergence of GDP per capita. It also requires that public services, income opportunities, and basic infrastructure be accessible to residents in different locations and that the gains from growth be shared between leading and lagging areas (World Bank, 2009; K. Zhang & Pei, 2019). In the Chinese context, RCD has been pursued through fiscal transfers, infrastructure investment, poverty alleviation, and, more recently, digitalization. The digital agenda is expected to reduce distance-related costs, expand market access, and improve the targeting of public services.

Digital factor allocation can affect RCD through several channels. First, digital infrastructure reduces communication and transaction costs, allowing firms in lagging regions to access customers, suppliers, and information that were previously available only in core regions (Czernich et al., 2011; Hjort & Poulsen, 2019). Second, digital finance extends payment, savings, credit, and insurance services to households and small firms. Mobile money and digital payments can improve risk sharing, smooth consumption, and relax liquidity constraints (Beck et al., 2007; Demirgüç-Kunt et al., 2022; Jack & Suri, 2014). Third, digital industry creates new tasks and occupations, and it can raise productivity in tradable services that are less dependent on physical distance (Goldfarb & Tucker, 2019; Huang et al., 2019). Fourth, digital human capital determines whether local workers and managers

can use digital tools productively; without complementary skills, access to technology may not translate into income or productivity gains (Akerman et al., 2015; van Dijk, 2005).

These channels suggest a positive relationship between DFA and RCD. The empirical literature provides support. Broadband infrastructure has been associated with economic growth and employment in both developed and developing countries (Czernich et al., 2011; Hjort & Poulsen, 2019). Digital finance has been linked to inclusive growth and poverty reduction (Beck et al., 2007; X. Zhang et al., 2019). Digitalization can also improve the efficiency of public services and the reach of social programs. At the same time, the literature cautions that the benefits are conditional on complementary factors. Forman et al. (2012) show that the internet's wage effects are concentrated in already prosperous areas, and Akerman et al. (2015) find that broadband complements skilled labor. These findings imply that DFA may not have a uniform effect across regions.

2.2. Absorptive Capacity and Threshold Effects

The idea that digitalization has stage-dependent effects is grounded in the absorptive-capacity literature. Digital technologies are general-purpose technologies: their productivity and welfare effects depend on how they are combined with organizational change, human capital, and institutional quality (Brynjolfsson & McAfee, 2014; O'Mahony & Vecchi, 2005). In regions with low income, thin markets, and weak public capacity, digital inputs may be used for consumption or communication rather than for productive investment. In regions with higher income and stronger complementary conditions, the same inputs can be combined with skills, finance, and infrastructure to generate larger coordination gains. This reasoning predicts that the marginal effect of DFA increases with economic development.

A threshold model is a natural way to test this prediction. Hansen (1999) develops a fixed-effects panel threshold estimator that allows the slope of a regressor to differ across regimes defined by an observed threshold variable. The estimator searches for the threshold that minimizes the residual sum of squares and uses bootstrap methods to test the null of linearity. The approach has been extended to instrumental variables and dynamic panels (Caner & Hansen, 2004; Seo & Shin, 2016), and it has been widely used to study nonlinear relationships in Chinese regional data. Qi et al. (2025), for example, find threshold effects of digital development, market efficiency, and technological progress on Chinese-style modernization. Their results suggest that the digital dividend is not automatic; it depends on the broader development environment.

The threshold interpretation also connects to the digital-divide literature. Access to digital infrastructure is only the first level of the divide; the second and third levels concern digital skills and the ability to convert access into material benefits (Norris, 2001; van Dijk, 2005; Warschauer, 2003). In low-income regions, the first-level divide may be closing, but the second- and third-level divides may persist. If so, the effect of DFA on RCD may be weak or even negative at low levels of development and stronger once complementary capabilities are in place. This motivates a threshold specification rather than a linear one.

2.3. Financial Development and Diminishing Returns

Digital finance is a core component of DFA, and its effects may depend on the depth of the financial system. In regions with limited financial development, digital finance can substitute for missing bank branches and expand access to basic financial services. In regions with deep financial systems, digital finance may add less at the extensive margin and may instead intensify competition, encourage speculative activity, or divert resources from productive investment. The relationship between financial depth and inclusion is

therefore unlikely to be linear (Beck et al., 2007). Beyond a certain level, additional financial deepening may have diminishing effects on coordination.

The Chinese experience is consistent with this possibility. Digital finance has expanded rapidly in leading cities, where traditional financial services were already relatively abundant, and more slowly in some rural areas. The marginal contribution of digital finance to RCD may therefore be larger where financial services are scarce and smaller where they are already plentiful. This does not imply that digital finance is harmful; rather, it suggests that the marginal effect may decline once a high level of financial development is reached. We therefore examine financial development as a second threshold variable and expect a smaller effect at very high levels of financial depth.

2.4. Regional Heterogeneity and Mechanisms

China's regions differ in initial digital endowments, industrial structure, and institutional capacity. The eastern region has a denser digital infrastructure, a larger share of high-skilled labor, and a more developed financial system. The central and western regions have lower initial levels of digitalization but may have more room for catch-up. The northeastern region has a legacy of heavy industry and a relatively slow digital transition. These differences imply that the effect of DFA on RCD may vary across regions. On the one hand, the absorptive-capacity argument predicts a stronger effect in the east, where complementary factors are more abundant. On the other hand, the catch-up argument predicts a stronger effect in the west, where the marginal return to digital infrastructure and finance may be higher. The empirical analysis can distinguish between these possibilities.

Three mechanisms are particularly relevant. First, DFA can promote industrial upgrading by reducing the cost of information, expanding market access, and enabling new service activities (Gan et al., 2011; Huang et al., 2019). Second, DFA can deepen financial development by expanding the coverage and use of digital financial services (Guo et al., 2020; X. Zhang et al., 2019). Third, DFA can increase technological investment by lowering the cost of innovation-related information and by creating demand for digital skills (Akerman et al., 2015). If these channels operate, DFA should be positively associated with industrial structure upgrading, financial development, and technological investment, and these variables should in turn be associated with RCD. The channel analysis in this study is exploratory, but it helps interpret the reduced-form threshold results.

2.5. Hypotheses

The discussion above yields five testable hypotheses.

H1. Digital factor allocation is positively associated with regional coordinated development.

H2. The effect of digital factor allocation on regional coordinated development is nonlinear; the marginal effect is larger after economic development crosses a threshold, consistent with absorptive capacity.

H3. The effect of digital factor allocation varies with financial development and tends to diminish at very high levels of financial depth.

H4. The effect of digital factor allocation is regionally heterogeneous, with a stronger average effect in less-developed western provinces.

H5. Digital factor allocation promotes regional coordinated development through industrial upgrading, financial deepening, and technological investment.

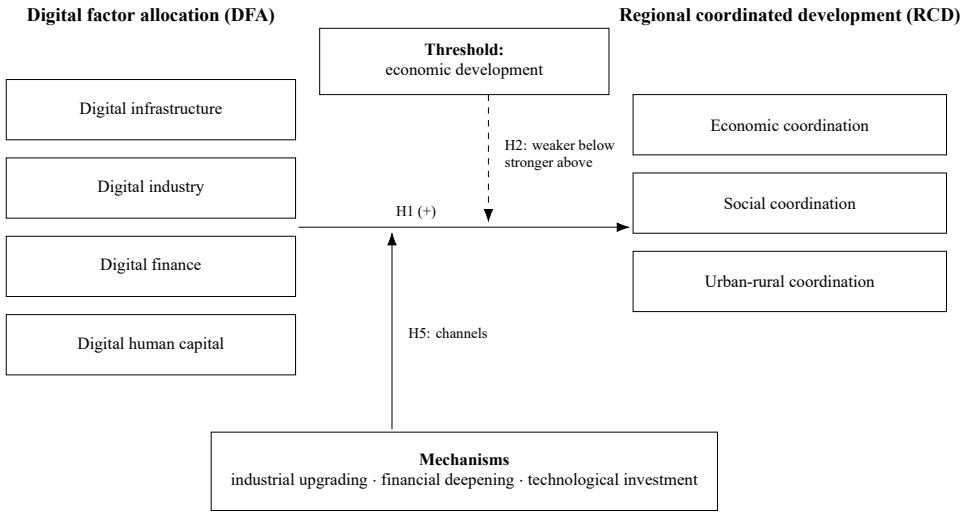


Figure 1. Research framework. DFA is measured across four dimensions and RCD across three dimensions. H1 tests the average association; H2 tests whether the DFA slope differs across economic-development regimes; H5 examines industrial upgrading, financial deepening, and technological investment as channels.

3. Data, Variables, and Methods

3.1. Study Area and Data

The analysis covers 30 provincial-level administrative units in mainland China from 2013 to 2021. Tibet, Hong Kong, Macao, and Taiwan are excluded because of data availability and comparability. The resulting panel is balanced, with 270 province-year observations. The sample starts in 2013 because the National Bureau of Statistics implemented a unified urban-rural household survey in that year, which makes the urban and rural income series comparable over the sample period.

The data come from several official sources. Provincial economic, social, infrastructure, and environmental indicators are taken from the *China Statistical Yearbook* (2014–2022), the *China City Statistical Yearbook* (2014–2022), the *China Environmental Statistics Yearbook*, and provincial statistical yearbooks (National Bureau of Statistics of China, 2022a, 2022b). The digital financial inclusion sub-indices are from the Peking University Digital Financial Inclusion Index (Guo et al., 2020; Peking University Digital Finance Research Center, 2021). A small number of missing values were completed by linear interpolation within provinces. For the urban and rural income series, we cross-checked the extracted values against the accounting identity linking total disposable income to urban and rural incomes and the urbanization rate; cells that violated the identity by a wide margin were corrected against the original yearbook tables.

3.2. Digital Factor Allocation Index

DFA is measured with a four-dimension indicator system (Table 1). The four dimensions capture distinct aspects of digital factor allocation. Digital infrastructure measures the physical and network foundations that determine whether digital services can reach firms and households. Digital industry measures the scale and market application of digital production. Digital finance measures the coverage, intensity, and digitalization of financial services. Digital human capital measures the supply of workers who can use digital technologies productively.

The 15 indicators are selected according to three criteria: relevance to the dimension, availability across all provinces and years, and consistency of statistical definitions. To avoid the criticism that a single composite index can be dominated by a few large provinces,

the indicators are expressed as per capita, per GDP, or per area ratios where appropriate. The DFA index assigns equal weights to the four dimensions and uses the entropy method to weight indicators within each dimension. This design gives each factor category the same ex ante importance while letting the data determine the relative weight of indicators inside each category.

3.3. Regional Coordinated Development Index

RCD is measured with a three-dimension indicator system covering economic coordination, social coordination, and urban-rural coordination (Table 1). Economic coordination is measured by GDP per capita, retail sales per capita, and fiscal self-sufficiency. These indicators capture the level and sustainability of local economic development. Social coordination is measured by disposable income per capita, hospital beds per 10,000 people, higher-education students per 100,000 people, and the share of social security spending in fiscal expenditure. These indicators capture income, health, education, and social protection. Urban-rural coordination is measured by the urbanization rate, the urban-rural income ratio (a negative indicator), and the urban registered unemployment rate (a negative indicator). These indicators capture the balance between urban and rural areas and the stability of urban employment.

The three dimensions receive equal weights, and the entropy method is used to weight indicators within each dimension. The baseline RCD index is therefore a weighted average of the three subsystem scores. In robustness checks, we also construct a five-dimension index that adds infrastructure and ecological coordination, a coupling coordination degree index for the three core dimensions, and a global entropy-weighted index using the ten core indicators. These alternatives allow us to examine whether the results depend on the choice of aggregation method.

3.4. Entropy Weight Method

The entropy weight method is used to construct the composite indices. The method assigns larger weights to indicators with greater dispersion, on the grounds that they carry more information for distinguishing provinces. For indicator j and province-year observation i, the raw value x_{ij} is first winsorized at the 1st and 99th percentiles and then normalized. For positive indicators,

$$z_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}, \tag{1}$$

and for negative indicators,

$$z_{ij} = \frac{\max_i x_{ij} - x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}. \tag{2}$$

A small constant (10^{-4}) is added to avoid zeros. The normalized values are converted into proportions $p_{ij} = z_{ij} / \sum_i z_{ij}$, and the entropy of indicator j is

$$e_j = -\frac{1}{\ln N} \sum_{i=1}^{N} p_{ij} \ln p_{ij}, \tag{3}$$

where N is the number of observations. The weight of indicator j is

$$w_j = \frac{1 - e_j}{\sum_k (1 - e_k)}. \tag{4}$$

Table 1. Indicator systems for digital factor allocation (DFA) and regional coordinated development (RCD).

Dimension	Indicator	Definition and unit	Sign
Panel A: Digital factor allocation (DFA)			
Digital infrastructure	Internet access ports per capita	Internet broadband access ports / year-end population (ports per person)	+
	Optical cable density	Long-distance optical cable length / land area (km/km^2)	+
	Mobile phone penetration	Mobile phone subscribers per 100 people	+
	IPv4 addresses per capita	IPv4 addresses / year-end population	+
	Domain names per capita	Registered domain names / year-end population	+
Digital industry	Software revenue / GDP	Software business revenue / GDP (%)	+
	E-commerce sales / GDP	E-commerce transaction value / GDP (%)	+
	Firms with e-commerce	Share of enterprises with e-commerce transactions (%)	+
	Websites per 100 firms	Number of websites per 100 enterprises	+
	Information firms per capita	Information enterprises / year-end population	+
Digital finance	DFI coverage breadth	Peking University Digital Financial Inclusion Index sub-index	+
	DFI usage depth	Peking University Digital Financial Inclusion Index sub-index	+
	DFI digitization	Peking University Digital Financial Inclusion Index sub-index	+
Digital human capital	ICT employment share	Employment in information transmission, computer services, and software / total urban employment (%)	+
	Internet penetration	Internet users / year-end population (%)	+
Panel B: Regional coordinated development (RCD)			
Economic coordination	GDP per capita	Regional GDP / year-end population (CNY/person)	+
	Retail sales per capita	Retail sales of consumer goods / year-end population (CNY/person)	+
	Fiscal self-sufficiency	Local fiscal revenue / local fiscal expenditure (%)	+
Social coordination	Disposable income per capita	All residents' per capita disposable income (CNY/person)	+
	Hospital beds	Hospital beds per 10,000 people	+
	Higher-education students	Higher-education students per 100,000 people	+
	Social security spending	Social security and employment expenditure / fiscal expenditure (%)	+
Urban-rural coordination	Urbanization rate	Urban resident population / year-end population (%)	+
	Urban-rural income ratio	Urban disposable income / rural disposable income	−
	Urban unemployment rate	Registered urban unemployment rate (%)	−

Notes: All indicators are compiled from the *China Statistical Yearbook*, the *China City Statistical Yearbook*, provincial statistical yearbooks, and the Peking University Digital Financial Inclusion Index. DFA assigns equal weights to the four dimensions and entropy weights to indicators within each dimension. RCD assigns equal weights to the three dimensions and entropy weights to indicators within each dimension.

The composite score for observation i is

$$S_i = \sum_j w_j z_{ij}. \tag{5}$$

For DFA and RCD, the entropy weights are computed within each dimension, and the dimension scores are averaged with equal weights. This two-step procedure prevents a dimension with many indicators from receiving disproportionate weight in the overall index.

3.5. Baseline and Threshold Models

The baseline specification is a two-way fixed-effects model:

$$RCD_{it} = \alpha_0 + \alpha_1 DFA_{it} + \gamma' X_{it} + \mu_i + \delta_t + \varepsilon_{it}, \tag{6}$$

where i indexes provinces, t indexes years, μ_i and δ_t are province and year fixed effects, X_{it} is a vector of controls, and ε_{it} is an error term clustered at the province level.

To test H2 and H3, we estimate a Hansen (1999) panel threshold model:

$$RCD_{it} = \beta_0 + \beta_1 DFA_{it} \cdot I(q_{it} \leq \gamma) + \beta_2 DFA_{it} \cdot I(q_{it} > \gamma) + \gamma' X_{it} + \mu_i + \varepsilon_{it}, \tag{7}$$

where q_{it} is the threshold variable, γ is the threshold value, and $I(\cdot)$ is the indicator function. The model allows the slope of DFA to differ between the low and high regimes. We estimate γ by grid search over the inner 80% of the threshold variable's distribution, which avoids small regimes at the tails. The null hypothesis of linearity is tested with a cluster wild bootstrap. The bootstrap generates $y^* = \hat{y} + \varepsilon^*$, where ε^* is the fitted residual multiplied by a province-level Rademacher sign, and re-estimates the threshold and the F statistic in each replication. The p-value is the fraction of bootstrap statistics that exceed the observed statistic. We also report the likelihood-ratio confidence interval for γ, using the critical value 7.35 for a 95% interval (Hansen, 1999, 2000).

The threshold variable is economic development (log GDP per capita) for H2 and financial development for H3. We also report specifications using DFA itself, technological investment, and private economy vitality as threshold variables. These additional specifications help assess whether the nonlinearity is specific to the economic-development threshold or reflects a broader pattern.

3.6. Controls and Descriptive Statistics

The controls are selected to capture local conditions that may affect RCD but are not part of the RCD index. Government intervention is measured by the share of local government expenditure in GDP. It captures the role of fiscal policy in public-service provision and infrastructure investment. Foreign direct investment is measured by the log of realized FDI per capita. It captures external openness and technology transfer. Population density is measured by the log of year-end population per square kilometer. It captures agglomeration and the cost of service delivery. Private economy vitality is measured by the number of private enterprises per 10,000 people. It captures the local business environment and market development. The baseline controls therefore cover the fiscal, external, spatial, and market dimensions of local conditions without directly entering the RCD index.

Table 2 reports the descriptive statistics. The DFA index has a mean of 0.259 and a standard deviation of 0.179, with a minimum of 0.007 and a maximum of 1.000. The RCD index has a mean of 0.357 and a standard deviation of 0.148, with a minimum of 0.083 and a maximum of 0.846. The urban-rural income ratio averages 2.53 and ranges from 1.84 to 3.56.

The large variation in both indices is important for the threshold analysis: it implies that the sample contains provinces at very different stages of digital development and regional coordination.

Table 2. Descriptive statistics.

Variable	N	Mean	SD	Min	Median	Max
DFA index	270	0.259	0.179	0.007	0.223	1.000
RCD index	270	0.357	0.148	0.083	0.333	0.846
RCD5 index	270	0.414	0.117	0.137	0.406	0.805
RCD coupling coordination	270	0.555	0.180	0.051	0.550	1.000
Urban-rural income ratio	270	2.533	0.361	1.842	2.474	3.556
Government intervention	270	26.292	11.122	10.501	23.207	75.344
Log FDI per capita	270	18.762	1.250	15.540	18.402	21.650
Log population density	270	5.469	1.289	2.068	5.659	8.282
Private economy vitality	270	191.904	142.191	41.889	155.839	813.382
Financial development	270	7.604	3.072	3.241	6.892	19.634
Technological investment	270	2.204	1.511	0.539	1.559	6.757
Industrial structure upgrading	270	140.959	74.470	66.533	122.565	524.401

Notes: The sample covers 30 provinces from 2013 to 2021 (N = 270). DFA is the digital factor allocation index; RCD is the regional coordinated development index; RCD5 is the five-dimension alternative; RCD-CCD is the coupling coordination degree of the three core dimensions. See Table 1 for definitions.

4. Results

4.1. Spatial and Temporal Patterns

Figure 2 maps the DFA and RCD indices for 2013 and 2021. Three patterns stand out. First, both indices increased over the sample period. The national average DFA rose from 0.109 in 2013 to 0.379 in 2021, and the average RCD rose from 0.298 to 0.520. The increase was not confined to a few leading provinces; most provinces moved into higher classes between 2013 and 2021.

Second, the spatial distribution is persistent. The eastern region has the highest DFA and RCD values throughout the period, followed by the central region, the northeastern region, and the western region. In 2021, the eastern averages were 0.406 for DFA and 0.557 for RCD, compared with 0.174 and 0.312 in the western region. The ranking of provinces is relatively stable, which is consistent with the view that digital and economic advantages are cumulative.

Third, the gap in DFA is larger than the gap in RCD. In 2013, the province with the highest DFA had a value more than 80 times that of the lowest, while the RCD range was much narrower. By 2021, the DFA gap had narrowed but remained substantial, whereas the RCD gap had changed less. This pattern suggests that digital factors diffuse faster than the broader conditions required for coordinated development, which motivates the threshold analysis below.

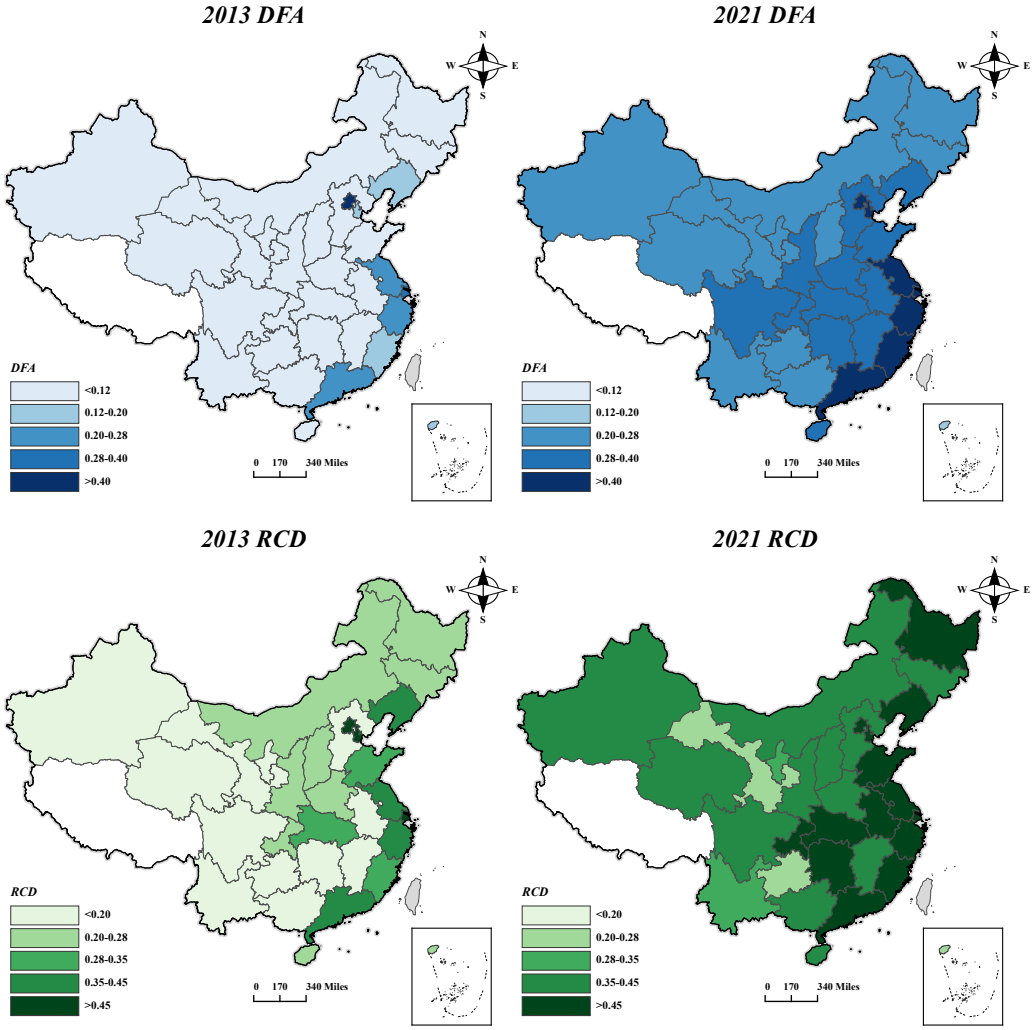


Figure 2. Spatial distribution of DFA and RCD, 2013 and 2021. (a) DFA; (b) RCD. Tibet is shown in white and Taiwan in light gray because they are not included in the estimation sample. The five classes use fixed cutoffs for comparability across years.

Figure 3 shows the regional averages over time. The eastern region maintains a clear lead in both indices. The western region starts from the lowest level but grows faster in DFA, especially after 2015. The northeastern region has a relatively high initial RCD but a slow digital transition, and its RCD growth is modest. These regional trajectories are consistent with the heterogeneity analysis in Section 4.4.

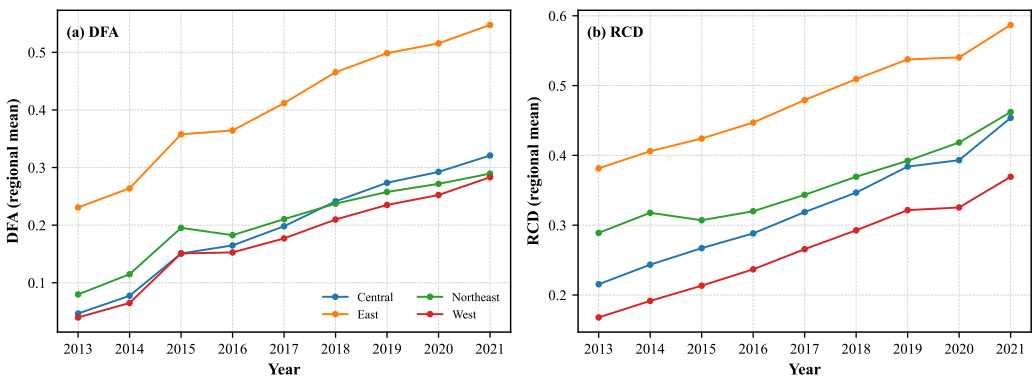


Figure 3. Regional averages of DFA and RCD, 2013–2021. (a) DFA; (b) RCD. Regions are defined as East, Central, West, and Northeast.

4.2. Baseline Estimates

Table 3 reports the baseline estimates. Column (1) includes DFA without controls and with province and year fixed effects. The coefficient is 0.277 and is significant at the 5% level. Column (2) adds the four controls. The coefficient falls to 0.241 but remains significant at the 5% level. Column (3) adds region-specific linear trends to absorb differential regional trends. The coefficient is 0.204 and is significant at the 10% level. Column (4) uses the five-dimension RCD index as the outcome; the coefficient is 0.013 and is not statistically significant.

The baseline estimate implies that a one-unit increase in the DFA index is associated with an increase of 0.241 points in the RCD index, or about 1.6 standard deviations of RCD. Since the DFA index ranges from 0 to 1, this is a large effect at the upper end of the distribution. A more modest increase of one standard deviation in DFA (0.179) is associated with an increase of about 0.043 points in RCD, or 0.29 standard deviations. The estimate is therefore economically meaningful as well as statistically significant.

The controls are generally insignificant. Government intervention, FDI, population density, and private economy vitality do not have robust associations with RCD once province and year fixed effects are included. This is not surprising: the controls are intended to absorb local conditions, not to explain RCD. Their inclusion does not change the DFA coefficient much, which suggests that the baseline result is not driven by these observable confounders.

The contrast between columns (2) and (4) is informative. DFA is positively associated with the three core coordination dimensions but not with the extended five-dimension index. The extended index adds infrastructure and ecological coordination, which are slower-moving and may respond to digitalization with a longer lag. The result suggests that the short-run effect of DFA operates mainly through economic, social, and urban-rural coordination, not through infrastructure or ecological outcomes. We return to this point in the robustness discussion.

Table 3. Baseline estimates of the effect of DFA on RCD.

	(1)	(2)	(3)	(4)
DFA	0.277**	0.241**	0.204*	0.013
	(0.112)	(0.118)	(0.118)	(0.133)
Government intervention		-0.0012	-0.0007	-0.0001
		(0.001)	(0.001)	(0.001)
Log FDI		0.0215	0.0169	0.0183
		(0.016)	(0.017)	(0.013)
Log population density		0.0794	0.1483	0.1003
		(0.138)	(0.203)	(0.129)
Private economy		-0.00002	0.00011	-0.00016
		(0.000)	(0.000)	(0.000)
Controls	No	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Region trends	No	No	Yes	No
Observations	270	270	270	270

Notes: The dependent variable is RCD in columns (1)–(3) and RCD5 in column (4). DFA is the digital factor allocation index. Controls include government intervention, log FDI per capita, log population density, and private economy vitality. Province and year fixed effects are included in all columns. Region trends are region-specific linear trends. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.3. Threshold Estimates

Table 4 reports the threshold tests and regime coefficients. Panel A shows the estimated threshold, the 95% likelihood-ratio confidence interval, the F statistic for the null of linearity, and the bootstrap p-value. Panel B shows the low- and high-regime coefficients of DFA.

The economic-development threshold is the main result. The estimated threshold is 10.589 in log GDP per capita, which corresponds to about CNY 39,700 per person. The bootstrap p-value is 0.003, so the null of linearity is rejected at the 1% level. The low-regime coefficient is 0.506 and the high-regime coefficient is 0.677; both are significant at the 1% level. The estimated marginal association is therefore about 34% larger after economic development crosses the threshold. Because the threshold model absorbs only province fixed effects, its coefficients are not directly comparable to the two-way fixed-effects estimates in Table 3; the relevant comparison is between the low and high regimes within the threshold model. This pattern is consistent with H2 and with the absorptive-capacity interpretation: digital factor allocation is more effective when local income, markets, and complementary capabilities are sufficiently developed.

Figure 4 plots the likelihood-ratio statistic for the economic-development threshold. The minimum is sharp, and the 95% confidence interval is narrow, ranging from 10.583 to 10.639. Figure 5 shows the raw association between DFA and RCD, split by the threshold. The high-development regime has a steeper slope, although the raw scatter also reflects the strong cross-sectional correlation between the two indices. The threshold estimate is therefore best read together with the fixed-effects estimates in Table 3, which absorb time-invariant province characteristics.

Table 4. Threshold tests and regime-specific coefficients.

Panel A: Threshold tests

Threshold variable	$\hat{\gamma}$	95% CI	F	Bootstrap p
DFA	0.178	[0.153, 0.349]	15.50	0.137
ln(per capita GDP)	10.589	[10.583, 10.639]	29.07	0.003
Financial development	10.045	–	61.48	0.010
Technological investment	0.856	[0.856, 0.998]	22.70	0.120
Private economy vitality	368.402	[85.480, 368.402]	23.81	0.107

Panel B: Regime-specific coefficients of DFA

Threshold variable	Low regime	High regime	Observations (low/high)
DFA	0.549***	0.661***	90/180
	(0.060)	(0.066)	
ln(per capita GDP)	0.506***	0.677***	66/204
	(0.057)	(0.067)	
Financial development	0.578***	0.267***	240/30
	(0.065)	(0.082)	
Technological investment	0.812***	0.627***	33/237
	(0.088)	(0.051)	
Private economy vitality	0.652***	0.573***	244/26
	(0.066)	(0.070)	

Notes: Each row corresponds to a separate threshold model with province fixed effects and the full control set. The threshold is estimated by grid search over the inner 80% of the threshold variable's distribution. The F statistic tests the null of a single linear regime. The bootstrap p-value is based on 300 cluster wild bootstrap replications. Standard errors clustered by province are in parentheses. For financial development, the likelihood-ratio interval collapses at the grid resolution; it is therefore not reported. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

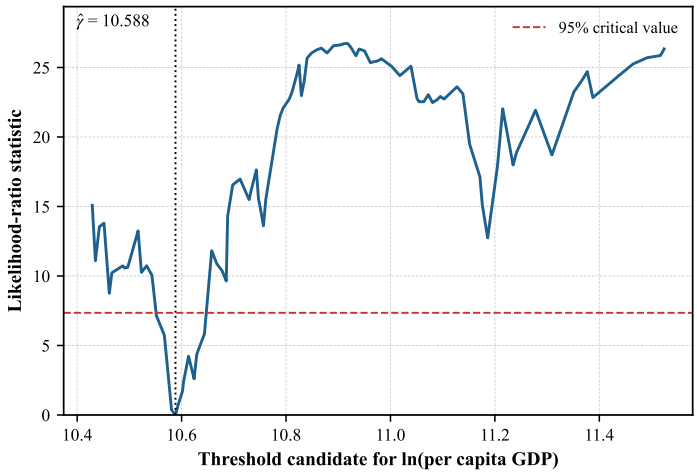


Figure 4. Likelihood-ratio statistic for the economic-development threshold. The dashed line is the 95% critical value (7.35). The dotted vertical line marks the estimated threshold.

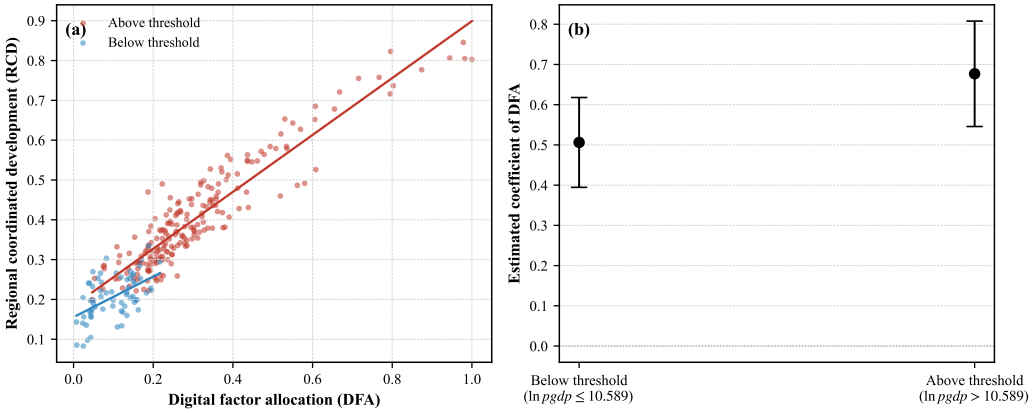


Figure 5. DFA, RCD, and the economic-development threshold. (a) Raw association between DFA and RCD, split by the estimated threshold; the lines are linear fits within each regime. (b) Regime-specific coefficients of DFA with 95% confidence intervals from the threshold model.

The financial-development threshold is also statistically significant, but its interpretation requires caution. The estimated threshold is 10.045. The low-regime coefficient is 0.578 and the high-regime coefficient is 0.267; the bootstrap p-value is 0.010. The high regime contains 30 observations: 27 from Beijing, Shanghai, and Tianjin, plus three observations from Qinghai. The decline in the marginal effect is therefore driven by a small group of financially deep municipalities. It is consistent with diminishing returns to digital finance, but it should not be read as a general statement about all high-finance provinces.

The other threshold variables do not produce statistically significant breaks. The DFA self-threshold has a bootstrap p-value of 0.137, the technology threshold 0.120, and the private-economy threshold 0.107. The point estimates suggest some variation across regimes, but the null of linearity cannot be rejected. The nonlinearity in the data is therefore primarily associated with economic development, not with the level of digitalization itself.

One caveat applies to all threshold estimates. The threshold model follows Hansen (1999) and includes province fixed effects but not year fixed effects. Because the threshold variable is a slow-moving development indicator, year fixed effects would absorb much of the cross-province variation in development stages that identifies the threshold. When region-specific linear trends are added, the economic-development threshold remains positive but becomes statistically weaker (bootstrap $p = 0.47$). The threshold should

therefore be interpreted as a long-run structural pattern rather than a short-run causal effect. We discuss this limitation further in Section 5.6.

4.4. Regional Heterogeneity

Table 5 reports the regional estimates. Under two-way fixed effects, the western region has a positive and highly significant coefficient of 0.555. The eastern and central coefficients are positive but insignificant (0.227 and 0.192), and the northeastern coefficient is negative and insignificant (−0.223). Under province fixed effects, all four regions have positive and significant coefficients, with the largest coefficients in the west (0.535) and central region (0.485), followed by the east (0.296) and the northeast (0.096).

The western result is the most robust. It holds under both specifications and is significant at the 1% level. This is consistent with H4 and with the catch-up interpretation: the marginal return to digital factor allocation is higher in regions that start with lower levels of digital infrastructure and financial access. The eastern and central results are sensitive to the inclusion of year fixed effects, which suggests that their average association is partly driven by common national trends. The northeastern result is weak and sometimes negative, which may reflect the region's industrial legacy and slow digital transition.

The regional pattern should not be overinterpreted. The subsample estimates are less precise than the full-sample estimates, and the differences across regions are not formally tested. The results are best read as descriptive evidence that the DFA–RCD relationship is not uniform across China's regions, with the west showing the most consistent positive association.

Table 5. Regional heterogeneity in the effect of DFA on RCD.

Region	Two-way FE	Province FE
East	0.227	0.296***
	(0.234)	(0.104)
Central	0.192	0.485***
	(0.340)	(0.072)
West	0.555***	0.535***
	(0.125)	(0.077)
Northeast	-0.223	0.096***
	(0.285)	(0.023)
Controls	Yes	Yes
Observations	90/54/99/27	90/54/99/27

Notes: Each cell reports the coefficient of DFA from a separate regression for the indicated region. Controls are included in all columns. Two-way FE includes province and year fixed effects; province FE includes province fixed effects only. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.5. Robustness

Table 6 reports the robustness checks. The baseline coefficient is 0.241. It remains positive and significant when the RCD index is replaced by a globally entropy-weighted index (0.349), when the DFA index is replaced by an equal-weight indicator index (0.298), when municipalities are excluded (0.288), when the sample ends in 2019 (0.283), and when the outcome and regressor are winsorized at the 1st and 99th percentiles (0.302). The coefficient is larger under province fixed effects (0.689) and smaller and insignificant when the five-dimension RCD index, the coupling coordination degree, the entropy-weighted DFA index, or lagged DFA are used.

The pattern is informative. The positive effect is robust to alternative weighting of the RCD indicators and to the exclusion of the largest municipalities. It is also robust to dropping 2020, which suggests that the COVID-19 shock is not driving the result. The effect is weaker for the five-dimension RCD index and for the coupling coordination degree. This is consistent with the baseline contrast in Table 3: the five-dimension index adds infrastructure and ecological dimensions that are less responsive to DFA in the short run, and the coupling coordination degree emphasizes balance among dimensions rather than levels.

The insignificant coefficient for lagged DFA suggests that the effect is contemporaneous rather than delayed. This may reflect the fast diffusion of digital services: once infrastructure and platforms are available, their coordination effects appear within the same year. It may also reflect measurement: the DFA index captures the current state of digital factor allocation, and its effect on RCD is likely to be strongest when the digital environment is contemporaneous with economic and social outcomes.

The entropy-weighted DFA index produces a positive but insignificant coefficient. This index places more weight on IPv4 addresses and domain names, which are concentrated in a few leading provinces. The equal-weight index and the baseline index, which give more balanced weight to infrastructure, industry, finance, and human capital, produce stronger results. This suggests that the effect is not driven by a single digital indicator but by the combined allocation of digital factors.

Table 6. Robustness checks.

Specification	Coefficient	SE	N
Baseline	0.241**	(0.118)	270
Five-dimension RCD	0.013	(0.133)	270
CCD of core dimensions	0.118	(0.288)	270
Entropy-weighted RCD	0.349**	(0.154)	270
Equal-weight DFA	0.298**	(0.143)	270
Entropy-weighted DFA	0.186	(0.176)	270
Excluding municipalities	0.288**	(0.128)	234
2013-2019	0.283**	(0.123)	210
Additional controls	0.213	(0.138)	270
Province FE only	0.689***	(0.069)	270
Lagged DFA	0.139	(0.146)	240
Winsorized 1/99	0.302**	(0.122)	270

Notes: Each row reports the coefficient of DFA (or lagged DFA) from a separate regression. The baseline includes the full control set and two-way fixed effects. The five-dimension RCD index adds infrastructure and ecological coordination. The coupling coordination degree is computed from the three core dimensions. The equal-weight DFA index uses equal weights for all 15 indicators. The entropy-weighted DFA index uses global entropy weights. The additional-controls specification adds technological investment, industrial structure upgrading, and financial development. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

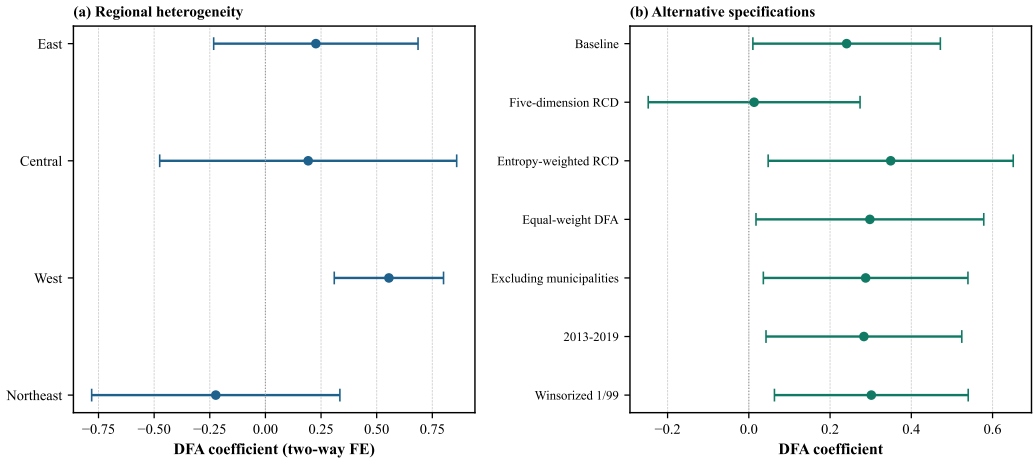


Figure 6. Regional and specification heterogeneity. (a) Regional DFA coefficients under two-way fixed effects. (b) DFA coefficients under alternative specifications. Bars are 95% confidence intervals.

4.6. Mechanisms

Table 7 reports the channel regressions. DFA is positively associated with technological investment (6.290, $p < 0.05$) and financial development (8.885, $p < 0.01$) under two-way fixed effects, and with industrial structure upgrading under province fixed effects (144.903, $p < 0.01$). These results are consistent with the view that digital factor allocation supports RCD through innovation, financial deepening, and structural change.

The income-gap result is more nuanced. Under two-way fixed effects, DFA is positively associated with the urban-rural income ratio (0.715, $p < 0.05$), which implies that a one-unit increase in DFA widens the urban-rural income gap in the short run. Under province fixed effects, the association is negative (-1.279, $p < 0.01$), which implies that over the longer run digital factor allocation is associated with a narrower gap. The contrast suggests that the distributional effect of digitalization is horizon-dependent: in the short run, digital opportunities may be captured disproportionately by urban and skilled groups, while in the longer run, the diffusion of digital services to rural areas may narrow the gap.

The mediator coefficients in the RCD equation are not statistically significant. This means that the channel evidence should be read as suggestive rather than as a formal mediation analysis. The reduced-form effect of DFA on RCD remains positive when the mediators are included, which suggests that the channels complement rather than fully explain the relationship. A more complete mediation analysis would require exogenous variation in each channel, which is beyond the scope of this study.

Table 7. Channel regressions.

Mediator	DFA → M (two-way FE)	DFA → M (province FE)	M in RCD equation
Technological investment	6.290**	1.255	0.006
	(2.985)	(1.275)	
Financial development	8.885***	4.941***	-0.001
	(3.164)	(1.773)	
Industrial upgrading	127.226	144.903***	-0.000
	(80.849)	(29.771)	
Urban-rural income ratio	0.715**	-1.279***	–
	(0.302)	(0.180)	
Controls	Yes	Yes	Yes
Observations	270	270	270

Notes: The first two columns report the effect of DFA on the mediator under two-way and province fixed effects, respectively. The third column reports the coefficient of the mediator in the RCD equation. Controls are included in all columns. Standard errors clustered by province are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

5. Discussion

5.1. Main Findings

This study set out to examine whether digital factor allocation promotes regional coordinated development and whether the relationship is nonlinear. The results provide a nuanced answer. On average, DFA is positively associated with RCD. The baseline two-way fixed-effects estimate is 0.241, and the association survives a range of robustness checks, including alternative weighting of the RCD index, the exclusion of municipalities, the exclusion of 2020, and winsorization. The effect is not uniform, however. It is stronger in provinces that have crossed a critical level of economic development, and it is concentrated in the western region. The threshold analysis also reveals a decline in the marginal effect at very high levels of financial development, although this result is driven by a small group of financially deep municipalities.

These findings contribute to the literature in three ways. First, they show that the digital dividend is conditional. The positive average effect does not mean that digital factor allocation is equally effective everywhere. The threshold results suggest that complementary economic conditions matter: digital inputs are more productive when they can be combined with skills, markets, and institutions. Second, the findings reconcile two apparently conflicting views in the literature. The digital-divide literature emphasizes that digitalization can widen gaps, while the digital-dividend literature emphasizes inclusion. Our results suggest that both can be true at different stages of development and over different time horizons. Third, the findings highlight the importance of measuring digital factors as a multidimensional construct. The composite DFA index has a positive effect, but the effect is not simply a reflection of digital infrastructure or digital finance alone.

5.2. Absorptive Capacity and the Digital Divide

The economic-development threshold is consistent with the absorptive-capacity explanation. Digital technologies are general-purpose technologies, and their effects depend on how they are combined with organizational change, human capital, and complementary infrastructure (Brynjolfsson & McAfee, 2014; O'Mahony & Vecchi, 2005). In low-income provinces, digital infrastructure may be used primarily for communication and consumption, while the productive and coordination effects require complementary inputs that are in short supply. In higher-income provinces, digital inputs can be combined with skilled labor, financial services, and market institutions to generate larger gains in economic, social, and urban-rural coordination.

This interpretation is also consistent with the multi-level digital divide. The first-level divide concerns access to infrastructure; the second-level divide concerns digital skills; and the third-level divide concerns the ability to convert access into material benefits (Norris, 2001; van Dijk, 2005; Warschauer, 2003). Our threshold result suggests that the third-level divide may be the binding constraint in low-income provinces. Providing access alone may not be enough; digital factor allocation needs to be accompanied by investments in education, public services, and market development.

The regional heterogeneity results reinforce this point. The western region has the strongest and most robust average effect, which is consistent with the catch-up interpretation: the marginal return to digital infrastructure and finance is higher where initial levels are low. At the same time, the threshold result suggests that the marginal effect is larger in economically developed provinces. These two findings are not necessarily inconsistent. The regional estimate compares average outcomes across regions over the sample period, while the threshold estimate compares provinces at different development stages within the same period. The west can have a larger average effect because it starts from a lower base, even if the marginal effect at a given level of development is smaller. The policy

implication is that lagging regions can benefit from digital factor allocation, but the size of the benefit depends on complementary conditions.

5.3. Financial Development and Diminishing Returns

The financial-development threshold suggests that the marginal effect of DFA declines at very high levels of financial depth. This is consistent with the idea that digital finance has diminishing returns at the extensive margin. In provinces with limited financial development, digital finance expands access to basic services and can have large coordination effects. In provinces with deep financial systems, digital finance adds less to access and may instead intensify competition or encourage speculative activity. The result should be interpreted with caution because the high-finance regime contains only 30 observations from Beijing, Shanghai, and Tianjin. These municipalities are not representative of all high-income provinces, and their digital and financial ecosystems are exceptional. The finding is therefore best read as a caution against assuming that more digital finance always produces more coordination, rather than as a general law.

The financial-development result also connects to the broader debate on financialization. Financial deepening can support growth and inclusion, but beyond a certain point it may divert resources from productive investment and increase fragility (Beck et al., 2007). Digital finance can amplify these tendencies because it lowers the cost of trading and expands the reach of financial products. The policy implication is not to restrict digital finance, but to ensure that it is directed toward productive and inclusive uses, with appropriate regulation and consumer protection.

5.4. Regional Heterogeneity and Inclusive Digitalization

The regional results show that the DFA–RCD association is strongest in the west and weaker or insignificant in the east, central, and northeast under two-way fixed effects. This pattern is consistent with the view that digital factor allocation can support inclusive digitalization when it reaches regions that were previously underserved. The west has a lower initial level of digital infrastructure and financial access, so the marginal effect of additional digital factors is larger. The east has already accumulated a high level of digital capital, so additional digital factors may have smaller effects on coordination, even if they raise productivity or innovation.

The northeastern result is weaker and sometimes negative. This may reflect the region's industrial structure and demographic challenges. The northeast has a relatively high initial level of urbanization and public services but a slow digital transition and a declining working-age population. Digital factor allocation may not be sufficient to offset these structural constraints. The result suggests that digital policies need to be embedded in a broader regional revitalization strategy that addresses industrial restructuring, human capital, and population outflow.

5.5. Policy Implications

These findings provide several practical implications for policymakers.

First, digital factor allocation should be sequenced with economic development. In low-income provinces, the priority should be basic connectivity, digital literacy, and complementary public services. Advanced digital industry and digital finance may have limited coordination effects until these foundations are in place. In higher-income provinces, the priority should shift toward deeper integration of digital technologies into production, public services, and urban-rural linkages.

Second, policies should be tailored to regional absorptive capacity. The western region has the strongest average effect, so it should be a priority for digital infrastructure and digital finance investment. However, investment should be paired with human capital

development, market reforms, and institutional capacity building. Without these complements, digital inputs may not translate into coordination gains.

Third, policymakers should address the short-run distributional effects of digitalization. The positive association between DFA and the urban-rural income ratio under two-way fixed effects suggests that urban and skilled groups may capture a disproportionate share of digital gains in the short run. Policies such as rural e-commerce, digital agriculture, digital public services, and targeted skills training can help distribute the gains more broadly. Over the longer run, the negative association under province fixed effects suggests that diffusion can narrow the gap, but the transition may be uneven.

Fourth, advanced regions should avoid over-digitalization and financialization. The financial-development threshold suggests that the marginal coordination effect of DFA declines at very high levels of financial depth. Advanced regions should focus on the quality and direction of digital investment rather than the quantity, strengthen regulation of digital finance, and ensure that digital innovation serves productive and inclusive purposes.

Fifth, the measurement of digital factors should be improved. The DFA index used here covers infrastructure, industry, finance, and human capital, but it does not capture data governance, platform competition, or the quality of digital public services. Future work can extend the index with indicators on data openness, digital security, and the integration of digital and physical infrastructure.

5.6. Limitations

Several limitations remain. First, the analysis is based on 30 provinces from 2013 to 2021. It does not capture within-province heterogeneity, which may be important in large provinces with divergent prefecture-level economies. City-level data could provide a more granular picture. Second, the DFA and RCD indices depend on the choice of indicators and weights. We address this partly by using alternative weighting schemes and alternative outcome definitions, but the indices are still simplifications of complex constructs.

Third, the threshold model identifies conditional associations, not causal effects. Digital factor allocation may be endogenous to regional coordination: provinces with stronger coordination may attract more digital investment, and unobserved shocks may affect both. The lagged-DFA specification does not produce a significant coefficient, which suggests that the relationship is contemporaneous and may be subject to reverse causality. Future research should exploit exogenous variation, such as historical telecommunications infrastructure, policy pilots, or terrain-based instruments, to identify causal effects.

Fourth, the threshold estimate is sensitive to the treatment of regional trends and year fixed effects. The baseline threshold model follows Hansen (1999) and includes province fixed effects only. Adding region-specific trends reduces the precision of the threshold estimate, and year fixed effects absorb much of the cross-province variation in development stages. The threshold should therefore be interpreted as a long-run structural pattern rather than a short-run causal effect. The financial-development threshold is based on a small high-regime group, which limits its generalizability.

Fifth, the income data and the urban-rural classification are subject to measurement issues. The household survey covers the resident population, while the urbanization rate is based on the resident population; the two are not perfectly consistent. We cross-checked the income series against the accounting identity, but some measurement error remains. The short-run and long-run differences in the income-gap results should be interpreted with this in mind.

Finally, the sample includes the COVID-19 period. The results are robust to dropping 2020, but 2021 may still be affected by the pandemic and by changes in digital adoption.

Future research can extend the sample and examine whether the threshold pattern persists in the post-pandemic period.

6. Conclusions

This study has examined the relationship between digital factor allocation and regional coordinated development in China. It constructed a four-dimension DFA index covering digital infrastructure, digital industry, digital finance, and digital human capital, and a three-dimension RCD index covering economic, social, and urban-rural coordination. Using a balanced panel of 30 provinces from 2013 to 2021 and a Hansen panel threshold model, the study tested whether the effect of DFA on RCD is linear and whether it depends on economic development and financial development.

The results show that DFA is positively associated with RCD. The baseline two-way fixed-effects estimate is 0.241, and the association remains positive and significant across several alternative specifications, including alternative weighting of the RCD index, the exclusion of municipalities, the exclusion of 2020, and winsorization. The effect is economically meaningful: a one-standard-deviation increase in DFA is associated with an increase of about 0.043 points in RCD, or 0.29 standard deviations.

The relationship is nonlinear. When per capita GDP exceeds the estimated threshold of 10.589 (about CNY 39,700), the marginal effect of DFA rises from 0.506 to 0.677. This pattern is consistent with absorptive capacity: digital factors are more productive when they are combined with sufficient income, markets, and complementary capabilities. A financial-development threshold indicates that the marginal effect declines at very high levels of financial depth, consistent with diminishing returns to digital finance, although this result is driven by a small group of financially deep municipalities.

The regional analysis shows that the effect is strongest in the western region, where initial levels of digital infrastructure and financial access are lower. This is consistent with the catch-up interpretation and with the view that digital factor allocation can support inclusive digitalization when it reaches underserved regions. The channel analysis suggests that industrial upgrading, financial deepening, and technological investment are plausible mechanisms. The income-gap results are horizon-dependent: DFA is associated with a wider urban-rural gap in the short run and a narrower gap in the long run under province fixed effects.

These findings have several policy implications. First, digital factor allocation should be sequenced with economic development: basic connectivity and digital literacy should come first in low-income provinces, while advanced digital integration should be the priority in higher-income provinces. Second, policies should be tailored to regional absorptive capacity, with the western region as a priority for digital infrastructure and digital finance investment, paired with human capital and market reforms. Third, policymakers should address the short-run distributional effects of digitalization through rural e-commerce, digital agriculture, digital public services, and targeted skills training. Fourth, advanced regions should avoid over-digitalization and financialization, focusing on the quality and direction of digital investment rather than the quantity.

Several limitations remain. The analysis is based on provincial data and does not capture within-province heterogeneity. The indices depend on indicator selection and weighting, and the threshold model identifies conditional associations rather than causal effects. The threshold estimate is sensitive to the treatment of regional trends and year fixed effects, and the financial-development threshold is based on a small high-regime group. Future research can address these limitations with city-level data, exogenous instruments, spatial and dynamic threshold models, and firm-level microdata. It can also extend the

DFA index to include data governance, digital security, and the quality of digital public services.

Overall, this study provides a multidimensional and nonlinear account of how digital factor allocation relates to regional coordinated development. The findings suggest that digitalization is not a universal remedy for regional disparities. Its benefits depend on the stage of economic development, the depth of the financial system, and the region's absorptive capacity. A differentiated policy approach that combines digital investment with complementary economic and social reforms is more likely to deliver inclusive and sustainable coordination gains.

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Appendix A Index Weights

Table A1 reports the within-dimension entropy weights for DFA and the global entropy weights for RCD. The DFA index assigns equal weights to the four dimensions. The RCD baseline assigns equal weights to the three dimensions; the weights below refer to the global entropy-weighted alternative used in the robustness checks.

Table A1. Entropy weights for DFA and RCD indicators.

Dimension	Indicator	Weight
Panel A: DFA (equal weights across four dimensions)		
Digital infrastructure	Internet access ports per capita	0.057
Digital infrastructure	Optical cable density	0.211
Digital infrastructure	Mobile phone penetration	0.056
Digital infrastructure	IPv4 addresses per capita	0.414
Digital infrastructure	Domain names per capita	0.262
Digital industry	Software revenue/GDP	0.336
Digital industry	E-commerce sales/GDP	0.214
Digital industry	Share of firms with e-commerce	0.085
Digital industry	Websites per 100 firms	0.042
Digital industry	Information firms per capita	0.323
Digital finance	DFI coverage breadth	0.303
Digital finance	DFI usage depth	0.376
Digital finance	DFI digitization	0.322
Digital human capital	ICT employment share	0.762
Digital human capital	Internet penetration	0.238
Panel B: RCD (global entropy weights)		
Economic	GDP per capita	0.157
Economic	Retail sales per capita	0.156
Economic	Fiscal self-sufficiency	0.084
Social	Disposable income per capita	0.150
Social	Hospital beds per 10,000	0.080
Social	Higher-education students per 100,000	0.073
Social	Social security spending share	0.102
Urban-rural	Urbanization rate	0.081
Urban-rural	Urban-rural income ratio (negative)	0.049
Urban-rural	Urban unemployment rate (negative)	0.066

Notes: Panel A reports within-dimension entropy weights for DFA; each of the four dimensions receives a weight of 0.25 in the composite. Panel B reports global entropy weights for the ten RCD indicators, used in the entropy-weighted RCD robustness check.

Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, *128*(6), 2188–2244. https://doi.org/10.1086/705716.

Akerman, A., Gaarder, I., & Mogstad, M. (2015). The skill complementarity of broadband internet. *Quarterly Journal of Economics*, *130*(4), 1781–1824. https://doi.org/10.1093/qje/qjv028.

Autor, D. H., & Salomons, A. (2018). Is automation labor-displacing? Productivity growth, employment, and the labor share. *Brookings Papers on Economic Activity*, *2018*(1), 1–87. https://doi.org/10.1353/eca.2018.0000.

Beck, T., Demirgüç-Kunt, A., & Levine, R. (2007). Finance, inequality and the poor. *Journal of Economic Growth*, *12*(1), 27–49. https://doi.org/10.1007/s10887-007-9010-6.

Brynjolfsson, E., & McAfee, A. (2014). *The second machine age: Work, progress, and prosperity in a time of brilliant technologies*. New York, NY, USA: W. W. Norton.

Caner, M., & Hansen, B. E. (2004). Instrumental variable estimation of a threshold model. *Econometric Theory*, *20*(5), 813–843. https://doi.org/10.1017/S0266466604205019.

Czernich, N., Falck, O., Kretschmer, T., & Woessmann, L. (2011). Broadband infrastructure and economic growth. *Economic Journal*, *121*(552), 505–532. https://doi.org/10.1111/j.1468-0297.2011.02420.x.

Demirgüç-Kunt, A., Klapper, L., Singer, D., & Ansar, S. (2022). *The global findex database 2021: Financial inclusion, digital payments, and resilience in the age of covid-19*. Washington, DC, USA: World Bank.

Forman, C., Goldfarb, A., & Greenstein, S. (2012). The internet and local wages: A puzzle. *American Economic Review*, *102*(1), 556–575. https://doi.org/10.1257/aer.102.1.556.

Gan, C., Zheng, R., & Yu, D. (2011). The impact of China's industrial structure changes on economic growth and fluctuations. *Economic Research Journal*, *46*(5), 4–16. (In Chinese)

Goldfarb, A., & Tucker, C. (2019). Digital economics. *Journal of Economic Literature*, *57*(1), 3–43. https://doi.org/10.1257/jel.20171452.

Guo, F., Wang, J., Wang, F., Kong, T., Zhang, X., & Cheng, Z. (2020). Measuring the development of China's digital financial inclusion: Index compilation and spatial characteristics. *China Economic Quarterly*, *19*(4), 1401–1418. (In Chinese)

Hansen, B. E. (1999). Threshold effects in non-dynamic panels: Estimation, testing, and inference. *Journal of Econometrics*, *93*(2), 345–368. https://doi.org/10.1016/S0304-4076(99)00025-1.

Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica*, *68*(3), 575–603. https://doi.org/10.1111/1468-0262.00124.

Hjort, J., & Poulsen, J. (2019). The arrival of fast internet and employment in Africa. *American Economic Review*, *109*(3), 1032–1079. https://doi.org/10.1257/aer.20161385.

Huang, Q., Yu, Y., & Zhang, S. (2019). Internet development and manufacturing productivity: Mechanisms and Chinese experience. *China Industrial Economics*(8), 5–23. (In Chinese)

Jack, W., & Suri, T. (2014). Risk sharing and transactions costs: Evidence from Kenya's mobile money revolution. *American Economic Review*, *104*(1), 183–223. https://doi.org/10.1257/aer.104.1.183.

National Bureau of Statistics of China. (2022a). *China city statistical yearbook 2014–2022.* (China Statistics Press: Beijing, China (in Chinese))

National Bureau of Statistics of China. (2022b). *China statistical yearbook 2014–2022.* (China Statistics Press: Beijing, China (in Chinese))

Norris, P. (2001). *Digital divide: Civic engagement, information poverty, and the internet worldwide*. Cambridge, UK: Cambridge University Press.

O'Mahony, M., & Vecchi, M. (2005). Quantifying the impact of ICT capital on output growth: A heterogeneous dynamic panel approach. *Economica*, *72*(288), 615–633. https://doi.org/10.1111/j.1468-0335.2005.00449.x.

Peking University Digital Finance Research Center. (2021). *Peking university digital financial inclusion index.* (Available online: https://www.idf.pku.edu.cn/ (accessed on 30 October 2026))

Qi, T., Liu, W., & Chang, X. (2025). Digital economy and Chinese-style modernization: Unveiling nonlinear threshold effects and inclusive policy frameworks for global sustainable development. *Economies*, *13*(8), 215. https://doi.org/10.3390/economies13080215.

Seo, M. H., & Shin, Y. (2016). Dynamic panels with threshold effect and endogeneity. *Journal of Econometrics*, *195*(2), 169–186. https://doi.org/10.1016/j.jeconom.2016.03.005.

van Dijk, J. A. G. M. (2005). *The deepening divide: Inequality in the information society*. Thousand Oaks, CA, USA: Sage.

Warschauer, M. (2003). *Technology and social inclusion: Rethinking the digital divide*. Cambridge, MA, USA: MIT Press.

World Bank. (2009). *World development report 2009: Reshaping economic geography*. Washington, DC, USA: World Bank.

Zhang, K., & Pei, X. (2019). Measurement of China's regional coordinated development level: An analysis based on provincial data. *Journal of Zhengzhou University (Philosophy and Social Sciences Edition)*, *52*(5), 55–61. (In Chinese)

Zhang, X., Wan, G., Zhang, J., & He, Z. (2019). Digital economy, inclusive finance, and inclusive growth. *Economic Research Journal*, *54*(8), 71–86. (In Chinese)

Zhao, T., Zhang, Z., & Liang, S. (2020). Digital economy, entrepreneurship, and high-quality development: Evidence from Chinese cities. *Journal of Management World*, *36*(10), 65–76. (In Chinese)